Sets And Functions

* Defn 1.1 $\mathbb{N}$: The set of natural numbers

$$\mathbb{N} := \{1, 2, 3 \ldots\}$$

** Well Ordering Property of $\mathbb{N}$: Every subset of $\mathbb{N}$ has a smallest element

* Defn 1.2 A set A is called a proper subset of B, ie $A \subsetneq B$, if $A \subset B$ and $A \neq B$.

* Defn 1.3 Set A and B are called disjoint if $A \cap B = \phi$

* Defn 1.4 Set theoretic difference of A and B, a.k.a the complement of B relative to A

$$A \setminus B = \{x : x \in A \text{ and } x \notin B\}$$

* Lemma 1.5

$$A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C)$$
$$A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$$

* Theorem 1.6 (Principle of Induction) Let $P(n)$ be a statement that depends upon the natural number n. Suppose that

(1) $P(1)$ is true

(2) If $P(n)$ is true then $P(n+1)$ is also true.

Then $P(n)$ is true $\forall n \in \mathbb{N}$

Hint: Proof by contradiction use 1.1

* Defn 1.7 The Cartesian product of two set A and B is defined as

$$A \times B := \{(x, y) : x \in A, y \in B\}$$

* Defn 1.8 A function $f : A \longrightarrow B$ is the subset of $A \times B$ s.t for each $x \in A$ $\exists$ a unique $y \in B$ for which $(x, y) \in f$.

$(x, y) \in f$ is written as $f(x) = y$

$\ast$ Defn 1.9

$$f: \overset{\text{Domain of } f}{\overline{A}} \longrightarrow \overset{\text{Codomain of } f}{\overline{B}}$$

Range of f $\quad R(f) := \{ y \in B : \exists x \in A \text{ s.t } f(x) = y \}$

Image / Direct Image of $C \subset A$ $\quad f(C) := \{ f(x) : x \in C \}$

Inverse image of $D \subset B$ $\quad f^{-1}(D) := \{ x \in A : f(x) \in D \}$

$\ast$ Lemma 1.10 $\quad f: A \to B$ and $C, D \subset B$ then

(i) $f^{-1}(C \cap D) = f^{-1}(C) \cap f^{-1}(D)$

(ii) $f^{-1}(C \cup D) = f^{-1}(C) \cup f^{-1}(D)$

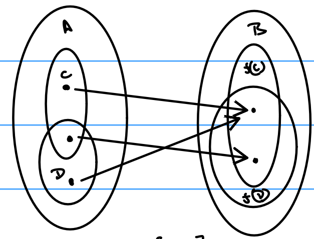
(iii) $f^{-1}(B \setminus C) = A \setminus f^{-1}(C)$

$\ast$ Lemma 1.11 $\quad f: A \to B$ and $C, D \subset A$ then

(i) $f(C \cup D) = f(C) \cup f(D)$

(ii) $f(C \cap D) \subset f(C) \cap f(D)$

Proof Hint



Example $\quad f(x) := x^2 \quad C = \{-2\} \quad D = \{2\}$

$\ast$ Defn 1.12 $\quad$ A binary relation on a set A is a subset $R \subset A \times A$ which contains the pairs where the relation holds. Instead of $(a, b) \in R$ we write aRb

$\ast$ Defn 1.13 $\quad$ Sets A and B have the same cardinality if there exists a bijective map $f: A \to B$

$\ast$ Defn 1.14 $\quad$ If a set A has the same cardinality as $\{1, 2, \dots, n\}$ for some $n \in \mathbb{N}$, we say $|A| := n$.

equivalence class of all set with same cardinality as A $\Big]$ finite cardinality

If A is empty we say $|A| := 0$

* Defn 1.15 $|A| \le |B|$ If $\exists$ a injective map $f : A \longrightarrow B$

* Defn 1.16 If $|A| = \mathbb{N}$, we say A is countably infinite

* Defn 1.17 If A is finite or countably infinite we say A is countable, else we say A is uncountable

* Defn 1.18 The power set of set A, denoted by $P(A)$, is the set of all subsets of A.

* Theorem 1.19 $|A| < P|A|$, ie there doesn't exist a surjective $f : A \longrightarrow P(A)$

Proof by contradiction: Let f be a surjective map

$S = \{x \in A : x \notin f(x)\}$, Let $c \in A$ s.t $f(c) = S$

Then either $c \in S$ or $c \notin S$

$\mathbb{R}$: Real Numbers

* Defn 2.1 An ordered set S is a set S together with a binary relation '<' s.t
(i) $\forall x, y \in S$, exactly one is true $<, >$ or $=$
(ii) If $x, y, z \in S$ s.t $x > y$ and $y > z$ then $x > z$

* Defn 2.2 Let $E \subset S$ where S is an ordered set
(i) if $\exists b \in S$ s.t $\forall x \in E$ $x \leq b$ then b is called an upper bound.
(ii) if $\exists b_0 \in S$ s.t (a) b_0 is an upper bound
(b) If b is another upper bound then $b_0 \leq b$
then b_0 is called the least upper bound or supremum of E
Similarly the greatest lower bound / infimum of E is defined.

* Defn 2.3 An ordered set S has the least upper bound property if every non empty set of S that is bounded above has a least upper bound.

* Defn 2.4 A set F is called field if it has two operations $(+)$ and $(\times)$ defined on it s.t
1. If $x, y \in F$ then $x+y \in F$
2. $\forall x, y, z \in F$ $(x+y)+z = x+(y+z)$
3. $\exists 0 \in F$ s.t $\forall x \in F$ $x+0 = 0+x = x$
4. $\forall x \in F$ $\exists -x \in F$ s.t $x+(-x) = -x+x = 0$

Abelian group under addition

5. $\forall x, y \in F$ $x+y = y+x$
6. If $x, y \in F$ then $x \cdot y \in F$
7. $\forall x, y, z \in F$ $(x \cdot y) \cdot z = x \cdot (y \cdot z)$
8. $\exists 1 \in F$ s.t $\forall x \in F$ $1 \cdot x = x \cdot 1 = x$
9. $\forall x \in \{F - \{0\}\}$ $\exists x^{-1} \in F$ s.t $x \cdot x^{-1} = x^{-1} \cdot x = 1$

Abelian group under multiplication

10. $\forall x, y \in F$ $x \cdot y = y \cdot x$

* Defn 2.5 An ordered field F is an ordered set s.t
(i) If $x, y, z \in F$ s.t $y < z$ then $x+y < x+z$
(ii) If $x > 0$ and $y > 0$ then $xy > 0$

*Lemma 2.6 Let F be an ordered field and $x, y, z, w \in F$ then

① If $x>0$ and $y>z$ then $xy > xz$

② If $x \neq 0$ then $x^2 > 0$ Hint $-1 \cdot x = -x$

③ $1>0$

④ $0<x<y$ then $x^2 < y^2$

⑤ $0<x<y$ then $0 < 1/y < 1/x$

*Lemma 2.7 An ordered field F has the least upper bound property if and only if it has the greatest lower bound property.

*Defn 2.8 Real Number, $\mathbb{R}$, is an unique ordered field with least upper bound property s.t $\mathbb{Q} \subset \mathbb{R}$.

*Lemma 2.9 If $S \subset \mathbb{R}$ is an upper bounded nonempty set, then $\forall \epsilon > 0$ $\exists$ $x \in S$ s.t

$$\sup(S) - \epsilon < x \leq \sup(S)$$

*Lemma 2.10 Let $A, B \subset \mathbb{R}$ be non empty subsets s.t $x \leq y$ $\forall x \in A$ and $y \in B$, then

$$\sup(A) \leq \inf(B)$$

*Theorem 2.11 If $x, y \in \mathbb{R}$ s.t $x>0$ and $x<y$ then $\exists n \in \mathbb{N}$ s.t $nx > y$.

(Archimedian property of $\mathbb{R}$)

Proof by contradiction: Rephrase $\mathbb{N} \subset \mathbb{R}$ is unbounded above
Use 2.9

*Theorem 2.12 $\mathbb{Q}$ is dense in $\mathbb{R}$ ie If $x, y \in \mathbb{R}$ st $x<y$ then

$\exists$ $r \in \mathbb{Q}$ s.t $x<r<y$

Proof Hint Let $x>0$ $\exists n \in \mathbb{N}$ s.t $n(y-x) \geq 1$. $A := \{k : k \in \mathbb{N} \text{ and } k > nx\}$

$m \in A$ be the smallest element of A. Then $m > nx \Rightarrow x < m/n$

Also $m-1 < nx \Rightarrow m < nx+1 \Rightarrow m < ny \Rightarrow m/n < y$

* Defn 2.13 Let $A \subset \mathbb{R}$ and $x \in \mathbb{R}$

$$x+A = \{x+y \in \mathbb{R} : y \in A\}$$
$$xA = \{xy \in \mathbb{R} : y \in A\}$$

* Lemma 2.14 If $A \subset \mathbb{R}$ is bounded above, then $\sup(x+A) = x + \sup(A)$

$\inf(x+A) = x + \inf(A)$

If $x > 0$, and A is bounded above $\sup(xA) = x\sup(A)$

$\inf(xA) = x\inf(A)$

* Defn 2.15 Let $A \subset \mathbb{R}$ be a set

(i) if A is empty then $\sup(A) := -\infty$

$\inf(A) := \infty$

(ii) If A is not bounded above then $\sup(A) := \infty$

(iii) " A " " " below then $\inf(A) := -\infty$

* Defn 2.16 When $A \subset \mathbb{R}$ is bounded above s.t. $\sup(A) \in A$ then we denote $\sup(A)$ by $\max(A)$. Similarly $\min(A)$ is defined

* Defn 2.17 Absolute value function $f : \mathbb{R} \longrightarrow \mathbb{R}^+$

$$|x| := \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

* Lemma 2.18

(i) $|x| \geq 0$ and $|x| = 0$ if and only if $x = 0$

(ii) $|xy| = |x|\,|y|$

(iii) $|x| \leq y$ if and only if $-y \leq x \leq y$

(iv) $-|x| \leq x \leq |x| \quad \forall x \in \mathbb{R}$

Hint: Divide it into different case based on signs of x and y

* Lemma 2.19 (Triangle Inequalities)

(i) $|x+y| \leq |x|+|y| \quad \forall x,y \in \mathbb{R}$

(ii) $|x-y| \leq |x|+|y| \quad$ "

(iii) $||x|-|y|| \leq |x-y| \quad$ "

* Defn 2.20 $f: D \to \mathbb{R}$. We say f is bounded above if $\exists M \in \mathbb{R}$ s.t $|f(x)| \leq M \quad \forall x \in D$

* Lemma 2.21 If $f: D \to \mathbb{R}$ and $g: D \to \mathbb{R}$ are bounded functions s.t $f(x) \leq g(x) \quad \forall x \in D$ then

$$\sup_{x\in D} f(x) \leq \sup_{x\in D} g(x) \qquad \inf_{x\in D} f(x) \leq \inf_{x\in D} g(x)$$

* Defn 2.22 Intervals

$$[a,b] := \{ x \in \mathbb{R} \text{ s.t } a \leq x \leq b\}$$
$$(a,b) := \{ x \in \mathbb{R} \text{ s.t } a < x < b\}$$

* Lemma 2.23 All intervals have the same cardinality ie $\exists$ a bijective map b/w any two intervals of $\mathbb{R}$

Example $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \xrightarrow{\tan(x)} \underbrace{(-\infty,\infty)}_{\mathbb{R}}$

* Theorem 2.24 $\mathbb{R}$ is uncountable

Proof Hint: Let $X \subset \mathbb{R}$ be countably infinite s.t if $a,b \in \mathbb{R}$ and $a<b$ then $\exists x \in X$ s.t $a<x<b$

$f: \mathbb{N} \to X$. Let $x_1 := f(1) \quad a_1 := x_1 \quad b_1 := x_1+1$

Construct $a_1 < a_2 \cdots < a_n < b_n < b_{n-1} \cdots < b_2 < b_1$ s.t $x_j \notin (a_k, b_k)$ where $j = 1,2,\dots,k$

$A = \{a_1, a_2, \dots, a_n\} \qquad B = \{b_1, b_2, \dots, b_n\}$

Let $y = \sup(A)$ then $y \notin A$, $y \notin B$ and $y \notin X$

$\therefore \mathbb{R}$ is uncountable

* Defn 2.25 For an infinite sequence of digits $0.d_1d_2d_3----$

$$D_n := \frac{d_1}{10} + \frac{d_2}{10^2} -- + \frac{d_n}{10^n}$$

* Lemma 2.26 (i) Every infinite sequence of digits $0.d_1d_2--$ represents unique $x \in [0,1]$ s.t

$$D_n \leq x \leq D_n + \frac{1}{10^n} \qquad \forall n \in \mathbb{N}$$

(ii) $\forall x \in (0,1]$ $\exists$ a unique infinite sequence of digits $0.d_1d_2d_3$ ---

that represents x s.t

$$D_n < x \leq D_n + \frac{1}{10^n} \qquad \forall n \in \mathbb{N}$$

Sequences And Series

* Defn 3.1 A sequence of real numbers, ie. $\{x_n\}_{n=1}^{\infty}$ is a function $f: \mathbb{N} \to \mathbb{R}$

* Defn 3.2 $\{x_n\}_{n=1}^{\infty}$ is called bounded if the underlying $\underbrace{\text{function is bounded}}_{\text{Defn 2.20}}$.

* Defn 3.3 $\{x_n\}_{n=1}^{\infty}$ is said to converge to $x \in \mathbb{R}$ if $\forall \varepsilon > 0 \; \exists M \in \mathbb{N}$ s.t $\forall n \geq M \quad |x_n - x| < \varepsilon$

Limit of the sequence $\lim_{n \to \infty} x_n = x$

* Defn 3.4 A sequence that is not convergent is called divergent.

* Lemma 3.5 A convergent sequence has a unique limit.
* Lemma 3.6 A convergent sequence is bounded.

* Lemma 3.7 If $\{x_n\}_{n=1}^{\infty}$ and $\{y_n\}_{n=1}^{\infty}$ are convergent series s.t $x_n \leq y_n \; \forall n \in \mathbb{N}$ then $\lim_{n \to \infty} x_n \leq \lim_{n \to \infty} y_n$

* Defn 3.8 $\{x_n\}_{n=1}^{\infty}$ is called monotone increasing if $x_{n+1} \geq x_n \quad \forall n \in \mathbb{N}$
" " " " decreasing " $x_{n+1} \leq x_n$ " "

* Theorem 3.9 A monotone sequence is bounded if and only it is convergent.
if $\{x_n\}_{n=1}^{\infty}$ is monotone increasing and bounded $\lim_{n \to \infty} x_n = \sup\{x_n : n \in \mathbb{N}\}$
" " " " decreasing " " $\lim_{n \to \infty} x_n = \inf\{x_n : n \in \mathbb{N}\}$

* Lemma 3.10 For $\{x_n\}_{n=1}^{\infty}$ the following statements are equivalent:
(i) $\{x\}_{n=1}^{\infty}$ is convergent
(ii) The K tail, ie $\{x_{n+K}\}_{n=1}^{\infty}$ converges $\forall K \in \mathbb{N}$
(iii) " " " " " converges for some $K \in \mathbb{N}$

* Defn 3.11 Let $\{n_i\}_{i=1}^{\infty}$ be a strictly increasing sequence of natural numbers. $\{x_{n_j}\}_{j=1}^{\infty}$ is called a subsequence of $\{x_n\}_{n=1}^{\infty}$

* Lemma 3.12 If $\{x_n\}_{n=1}^{\infty}$ is a convergent sequence then all its subsequences $\{x_{n_i}\}_{i=1}^{\infty}$ also convergent to the same limit

* Squeeze lemma 3.13 : If $\{x_n\}_{n=1}^{\infty}$, $\{a_n\}_{n=1}^{\infty}$ and $\{b_n\}_{n=1}^{\infty}$ are sequences and $L \in \mathbb{R}$ s.t

(i) $\{a_n\}_{n=1}^{\infty}$ and $\{b_n\}_{n=1}^{\infty}$ are convergent sequences

(ii) $\lim_{n\to\infty} a_n = \lim_{n\to\infty} b_n = L$

(iii) $a_n \le x_n \le b_n \quad \forall n \in \mathbb{N}$

then $\{x_n\}_{n=1}^{\infty}$ also converges to L

* Lemma 3.14 Let $\{x_n\}_{n=1}^{\infty}$, $\{y_n\}_{n=1}^{\infty}$ be convergent sequences then

(i) $\{z_n\}_{n=1}^{\infty}$ where $z_n := x_n + y_n$ converges to $\lim_{n\to\infty} z_n = \lim_{n\to\infty} x_n + \lim_{n\to\infty} y_n$

(ii) $\{z_n\}_{n=1}^{\infty}$ where $z_n := x_n y_n$ converges to $\lim_{n\to\infty} z_n = \lim_{n\to\infty} x_n \lim_{n\to\infty} y_n$

Proof Hint: Let $\varepsilon > 0$. $K = \max(|x|, |y|, \varepsilon/3, 1)$ $\quad \exists M_1 \in \mathbb{N}$ s.t $\forall n \ge M_1$ $|x_n - x| < \varepsilon/3K$

$\exists M_2 \in \mathbb{N}$ st $\forall n \ge M_2$ $|y_n - y| < \varepsilon/3K$

$\forall n \ge \max(M_1, M_2)$ Prove $|x_n y_n - xy| = |(x_n - x + x)(y_n - y + y) - xy| < \varepsilon$

(iii) if $y_n \neq 0$ $\forall n \in \mathbb{N}$ and $\lim_{n\to\infty} y_n \neq 0$ Then $\{z_n\}_{n=1}^{\infty}$ where $z_n := x_n / y_n$

converges to $\lim_{n\to\infty} z_n = \lim_{n\to\infty} x_n / \lim_{n\to\infty} y_n$

Proof Hint: Prove $\{1/y_n\}_{n=1}^{\infty}$ converges to $1/y$ where $y = \lim_{n\to\infty} y_n$

Let $\varepsilon > 0$ $\quad \exists M_1 \in \mathbb{N}$ st $|y_n - y| < \min\left(\frac{|y|^2 \varepsilon}{2}, \frac{|y|}{2}\right)$

$\forall n \ge M_1$ $\quad |y| = |y - y_n + y_n| \le |y - y_n| + |y_n| < \frac{|y|}{2} + |y_n| \Rightarrow \frac{1}{|y_n|} < \frac{2}{|y|}$

$$\left|\frac{1}{y_n} - \frac{1}{y}\right| = \left|\frac{y - y_n}{y y_n}\right| = \frac{|y - y_n|}{|y|\,|y_n|} < \frac{|y|^2 \varepsilon/2}{|y|} \times \frac{2}{|y|} < \varepsilon$$

* Lemma 3.15 If $\{x_n\}_{n=1}^{\infty}$ is a convergent sequence s.t $x_n \geq 0 \quad \forall n \in \mathbb{N}$, then

(i) $$\lim_{n\to\infty} \sqrt{x_n} = \sqrt{\lim_{n\to\infty} x_n}$$

(ii) $$\lim_{n\to\infty} |x_n| = |\lim_{n\to\infty} x_n|$$

* Lemma 3.16 Let $\{x_n\}_{n=1}^{\infty}$ be a sequence, $x \in \mathbb{R}$ and $\{a_n\}_{n=1}^{\infty}$ be another sequence s.t

(i) $\{a_n\}_{n=1}^{\infty}$ is convergent

(ii) $\lim_{n\to\infty} a_n = 0$

(iii) $|x_n - x| < a_n \quad \forall n \in \mathbb{N}$

then $\{x_n\}_{n=1}^{\infty}$ is convergent and $\lim_{n\to\infty} x_n = x$

* Lemma 3.17 Let $c > 0$

(i) If $c < 1$, the sequence $\{c^n\}_{n=1}^{\infty}$ converges to 0

(ii) If $c > 1$, " " " is unbounded

* Lemma 3.18 Let $\{x_n\}_{n=1}^{\infty}$ be a sequence s.t

(i) $x_n \neq 0 \quad \forall n \in \mathbb{N}$

(ii) $L = \lim_{n\to\infty} \frac{|x_{n+1}|}{|x_n|}$ exists

then (a) if $L < 1$ then the sequence $\{x_n\}_{n=1}^{\infty}$ converges to 0

(b) " $L > 1$ " " " " is unbounded

* Defn 3.19 Let $\{x_n\}_{n=1}^{\infty}$ be a bounded sequence. Define $\{a_n\}_{n=1}^{\infty}$ and $\{b_n\}_{n=1}^{\infty}$ sequences where $a_n := \sup\{x_k : k \geq n\}$ $\quad b_n := \inf\{x_k : k \geq n\}$

Limit superior $\limsup_{n\to\infty} x_n := \lim_{n\to\infty} a_n$

Limit inferior $\liminf_{n\to\infty} x_n = \lim_{n\to\infty} b_n$

*Lemma 3.20 For a bounded sequence $\{x_n\}_{n=1}^{\infty}$

$$\limsup_{n\to\infty} x_n := \lim_{n\to\infty} a_n = \inf\{a_n : n\in\mathbb{N}\}$$

$$\liminf_{n\to\infty} x_n := \lim_{n\to\infty} b_n = \sup\{b_n : n\in\mathbb{N}\}$$

Also $\liminf_{n\to\infty} x_n \le \limsup_{n\to\infty} x_n$

*Theorem 3.21 If $\{x_n\}_{n=1}^{\infty}$ is a bounded sequence, then $\exists$ subsequences $\underbrace{\{x_{n_i}\}_{i=1}^{\infty}, \{x_{m_j}\}_{j=1}^{\infty}}_{\text{Need not to be monotone}}$ s.t

$$\limsup_{n\to\infty} x_n = \lim_{i\to\infty} x_{n_i}$$

$$\liminf_{n\to\infty} x_n = \lim_{j\to\infty} x_{m_j}$$

Proof Hint: Let $n_1 = 1$ Let $n_1, n_2, \ldots, n_{k-1}$ are already defined

$m\in\mathbb{N}$ s.t m s.t $\forall n \ge m$ $|a_{n_{k-1}+1} - x_n| < 1/k$ Define $n_k = m$

then $\forall k \ge 2$ $|a_{n_k} - x_{n_k}| < \frac{1}{k}$ as $a_{n_k} \le a_{n_{k-1}+1}$

$\{a_{n_k}\}_{k=1}^{\infty}$ is subsequence of $\{a_n\}$ $\Rightarrow$ $\lim_{n\to\infty} a_n = \lim_{k\to\infty} a_{n_k} = a$

Let $\varepsilon > 0$, $\exists M_1 \in\mathbb{N}$ s.t $\forall i \ge M_1$ $|a_{n_i} - a| < \varepsilon/2$

$M_2\in\mathbb{N}$ s.t $1/M_2 < \varepsilon/2$ $j = \max(M_1, M_2, 2)$

$\forall i \ge j$ $|x_{n_i} - a| \le |x_{n_i} - a_{n_i}| + |a - a_{n_i}| < \frac{1}{i} + \varepsilon/2 < \frac{1}{j} + \varepsilon/2 \le \frac{1}{M_2} + \varepsilon/2 < \varepsilon/2 + \varepsilon/2 = \varepsilon$

*Theorem 3.22 Let $\{x_n\}_{n=1}^{\infty}$ be a bounded sequence. Then $\{x_n\}_{n=1}^{\infty}$ converges if and only if

$$\liminf_{n\to\infty} x_n = \limsup_{n\to\infty} x_n$$

Proof Hint: Use 3.21, 3.11
Use 3.13

Further $\liminf_{n\to\infty} x_n = \limsup_{n\to\infty} x_n = \lim_{n\to\infty} x_n$

*Lemma 3.23 If $\{x_n\}_{n=1}^{\infty}$ is a bounded sequence and $\{x_{n_k}\}_{k=1}^{\infty}$ is a subsequence, then

$$\liminf_{n\to\infty} x_n \le \liminf_{k\to\infty} x_{n_k} \le \limsup_{k\to\infty} x_{n_k} \le \limsup_{n\to\infty} x_n$$

* Lemma 3.24 A bounded sequence $\{x_n\}_{n=1}^{\infty}$ is convergent to x if and only if every convergent subsequence $\{x_{n_k}\}_{k=1}^{\infty}$ converges to x

Use 3.17, 3.21 and 3.18

* Theorem 3.25 Bolzano-Weierstrass: Suppose a sequence $\{x_n\}_{n=1}^{\infty}$ of real numbers is bounded. Then there exists a convergent subsequence $\{x_{n_k}\}_{k=1}^{\infty}$

* Defn 3.26 $\{x_n\}_{n=1}^{\infty}$ diverges to infinity, ie. $\lim\limits_{n\to\infty} x_n = \infty$ if $\forall K \in \mathbb{R}$ $\exists M \in \mathbb{N}$ s.t $\forall n \geq M$ $x_n > K$

$\{x_n\}_{n=1}^{\infty}$ diverges to $-\infty$, ie. $\lim\limits_{n\to\infty} x_n = -\infty$ if $\forall K \in \mathbb{R}$ $\exists M \in \mathbb{N}$ s.t $\forall n \geq M$ $x_n < K$

* Defn 3.27 Extension of defn of lim sup and lim inf to unbounded sequences

* Defn 3.28 A sequence $\{x_n\}_{n=1}^{\infty}$ is called a Cauchy sequence. If $\forall \epsilon > 0$, $\exists$ an $M \in \mathbb{N}$ s.t $\forall k, m \geq M$ $|x_m - x_k| < \epsilon$

* Lemma 3.29 If a sequence is Cauchy, it is bounded

* Theorem 3.30 A sequence of real numbers is Cauchy if and only if it is convergent.

Proof Hint: Cauchy $\Rightarrow$ bounded $\Rightarrow$ $\exists$ $\{x_{n_i}\}_{i=1}^{\infty}$ s.t $\lim\limits_{i\to\infty} x_{n_i} = \limsup\limits_{n\to\infty} x_n = a$ (Using 3.21)

$\hookrightarrow$ $\exists$ $\{x_{m_j}\}_{j=1}^{\infty}$ s.t $\lim\limits_{j\to\infty} x_{m_j} = \liminf\limits_{n\to\infty} x_n = b$

$\exists M_1 \in \mathbb{N}$ s.t $\forall i \geq M_1$ $|x_{n_i} - a| < \epsilon/3$

" $M_2 \in \mathbb{N}$ " $\forall j \geq M_2$ $|x_{m_j} - b| < \epsilon/3$

" $M_3 \in \mathbb{N}$ " $\forall i \geq M_3$ $|x_{n_i} - x_{m_j}| < \epsilon/3$

Let $i = \text{Max}(M_1, M_2, M_3)$

$|a-b| = |a - x_{n_i} + x_{n_i} - x_{m_i} + x_{m_i} - b|$

$< |x_{n_i} - a| + |x_{m_i} - b| + |x_{n_i} - x_{m_i}|$

$< \epsilon/3 + \epsilon/3 + \epsilon/3 < \epsilon$

$\therefore a = b$

Use 3.13

* Def^n 3.31 Given a sequence $\{x_n\}_{n=1}^{\infty}$, a series is defined as

$$\sum_{n=1}^{\infty} x_n .$$

A series is said to converge if the sequence of partial sums $\{s_k\}$ converges where

$$s_k := \sum_{n=1}^{k} x_n = x_1 + x_2 - - + x_k$$

* Lemma 3.32 (k-tail) Let $\sum_{n=1}^{\infty} x_n$ be a series and $M \in \mathbb{N}$, then $\sum_{n=1}^{\infty} x_n$ converges if and only if $\sum_{n=M}^{\infty} x_n$ converges

* Def^n 3.33 A series $\sum_{n=1}^{\infty} x_n$ is called Cauchy if the sequence of partial sums $\{s_k\}$ is Cauchy.

* Lemma 3.34 The series $\sum_{n=1}^{\infty} x_n$ is Cauchy if and only if $\forall \varepsilon > 0$ $\exists$ an $M \in \mathbb{N}$ s.t $\forall n \geq M$ and every $k > n$

$$\left|\sum_{i=n+1}^{k} x_i\right| < \varepsilon$$

* Lemma 3.35 Let $\sum_{n=1}^{\infty} x_n$ be a convergent series then the sequence $\{x_n\}_{n=1}^{\infty}$ is convergent to 0, ie

$$\lim_{n\to\infty} x_n = 0$$

* Lemma 3.36 Let $\alpha \in \mathbb{R}$ and $\sum_{n=1}^{\infty} x_n, \sum_{n=1}^{\infty} y_n$ be convergent series then

(i) $\sum_{n=1}^{\infty} \alpha x_n$ is also a convergent series and further

$$\sum_{n=1}^{\infty} \alpha x_n = \alpha \sum_{n=1}^{\infty} x_n$$

(ii) $\sum_{n=1}^{\infty} (x_n + y_n)$ is a convergent series and further

$$\sum_{n=1}^{\infty} (x_n + y_n) = \sum_{n=1}^{\infty} x_n + \sum_{n=1}^{\infty} y_n$$

* Def^n 3.37 Absolute convergence: A series $\sum_{n=1}^{\infty} x_n$ is said to converge absolutely if the series $\sum_{n=1}^{\infty} |x_n|$ converges.

A series is called conditionally convergent if it is convergent but not absolutely convergent.

* Lemma 3.38 If a series $\sum_{n=1}^{\infty} x_n$ is absolutely convergent then it is also convergent

* Lemma 3.39 (Comparison test) Let $\sum_{n=1}^{\infty} x_n$ and $\sum_{n=1}^{\infty} y_n$ are series s.t $0 \leq x_n \leq y_n$ $\forall n \in \mathbb{N}$ then

(i) If $\sum_{n=1}^{\infty} y_n$ is convergent then $\sum_{n=1}^{\infty} x_n$ is also convergent

Use 3.9, 3.6

(ii) If $\sum_{n=1}^{\infty} x_n$ diverges then $\sum_{n=1}^{\infty} y_n$ also diverges.

* Lemma 3.40 The series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ where $p \in \mathbb{R}$ is convergent if and only if $p < 1$

* Lemma 3.41 (Ratio test) Let $\sum_{n=1}^{\infty} x_n$ is a series s.t $x_n \neq 0$ $\forall n \in \mathbb{N}$ and $\exists$ $L = \lim_{n \to \infty} \frac{|x_{n+1}|}{|x_n|}$ then

(i) If $L < 1$ then $\sum_{n=1}^{\infty} x_n$ converges absolutely

Proof Hint: Use 3.18(a)

(ii) If $L > 1$ then $\sum_{n=1}^{\infty} x_n$ diverges

Continuous Functions

* Defn 4.1 Let $S \subset \mathbb{R}$. Then $x \in \mathbb{R}$ is called a cluster point of S is $\forall\, \varepsilon > 0$ the set $(x-\varepsilon, x+\varepsilon) \cap S/\{x\}$ is non empty

Rephrasing $\forall\, \varepsilon > 0$ $\exists\, y \in S$ s.t $y \neq x$ and $|x-y| < \varepsilon$

* Lemma 4.2 Let $S \subset \mathbb{R}$ and $x \in \mathbb{R}$ then x is a cluster point of S $\iff$ $\exists\, \{x_n\}_{n=1}^{\infty}$ s.t $\forall n$ $x_n \in S/\{x\}$ and $\lim_{n\to\infty} x_n = x$

Proof Hint use 3.16 with $a_n := 1/n$

* Defn 4.3 Let $S \subset \mathbb{R}$, $c \in \mathbb{R}$ be a cluster point of S, $f : S \longrightarrow \mathbb{R}$, $L \in \mathbb{R}$.

We say $\lim_{x\to c} f(x) = L$ or $f(x) \longrightarrow L$ as $x \longrightarrow c$

if $\forall\, \varepsilon > 0$, $\exists$ an $\delta > 0$ s.t

if $x \in S/\{c\}$ and $|x-c| < \delta \Rightarrow |f(x) - L| < \varepsilon$

* Lemma 4.4 The limit of a function $f(x)$ as $x \to c$ if it exists is unique

* Lemma 4.5 (Sequential Limit) Let $S \subset \mathbb{R}$ and c be its cluster point. Let $f : S \longrightarrow \mathbb{R}$ be a function.

$\lim_{x\to c} f(x) = L$ $\iff$ <u>for every</u> $\{x_n\}_{n=1}^{\infty}$ s.t $\forall\, n \in \mathbb{N}$ $x_n \in S/\{c\}$ and $\lim_{n\to\infty} x_n = c$, $\{f(x_n)\}_{n=1}^{\infty}$ <u>converges to L</u>.

Proof Hint: step 1 If A is true B is true

step 2 If $\underbrace{\text{A is false}}$ B is false

$\exists$ an $\varepsilon > 0$ s.t $\forall\, \delta > 0$ $\exists\, x \in S/\{x\}$ s.t $|x-c| < \delta$ and $|f(x) - L| \geq \varepsilon$

* Corollary 4.6 Let $S \subset \mathbb{R}$, c be a cluster point of S. Suppose $f : S \longrightarrow \mathbb{R}$ and $g : S \longrightarrow \mathbb{R}$ be functions s.t $f(x) \leq (x)$ $\forall\, x \in S$, and $\exists$ $\lim_{x\to c} f(x)$ and $\lim_{x\to c} g(x)$ then

$$\lim_{x\to c} f(x) \leq \lim_{x\to c} g(x)$$

Proof Hint:- Use 4.2, 4.5, 3.7

* Corollary 4.7 Let $S \subset \mathbb{R}$ and c be a cluster point of S. Suppose $f: S \to \mathbb{R}$, $g: S \to \mathbb{R}$, $h: S \to \mathbb{R}$ are functions s.t $g(x) \leq f(x) \leq h(x)$ $\forall x \in S$.

Suppose $\lim_{x \to c} g(x)$ and $\lim_{x \to c} h(x)$ exist and $\lim_{x \to c} g(x) = \lim_{x \to c} h(x)$ then

$$\lim_{x \to c} f(x) = \lim_{x \to c} g(x) = \lim_{x \to c} h(x)$$

Proof Hint: Use 4.2, 4.5, 3.13

Corollary 4.8 Let $S \subset \mathbb{R}$, c be a cluster point of S. Suppose $f: S \to \mathbb{R}$, $g: S \to \mathbb{R}$ be functions and $\lim_{x \to c} f(x)$ and $\lim_{x \to c} g(x)$ exists. Then

(i) $\lim_{x \to c} (f(x) + g(x)) = \left(\lim_{x \to c} f(x)\right) + \left(\lim_{x \to c} g(x)\right)$

(ii) $\lim_{x \to c} (f(x) \cdot g(x)) = \left(\lim_{x \to c} f(x)\right)\left(\lim_{x \to c} g(x)\right)$

Proof Hint: Use 4.2, 4.5, 3.14

(iii) $g(x) \neq 0$ $\forall x \in S / \{c\}$ and $\lim_{x \to c} g(x) \neq 0$

$$\lim_{x \to c} \left(\frac{f(x)}{g(x)}\right) = \frac{\lim_{x \to c} f(x)}{\lim_{x \to c} g(x)}$$

Corollary 4.9 Let $S \subset \mathbb{R}$, c be a cluster point of S. Suppose $f: S \to \mathbb{R}$ is a function s.t as $x \to c$ the limit of $f(x)$ exists. Then

$$\lim_{x \to c} |f(x)| = \left|\lim_{x \to c} f(x)\right|$$

Use 4.2, 4.5, 3.15

* Defn 4.10 (Restriction of f to A) Let $f: S \to \mathbb{R}$ and $A \subset S$. Define the function $f|_A : A \to \mathbb{R}$ by

$$f|_A(x) := f(x) \quad \text{for } x \in A$$

* **Lemma 4.11** Let $S \subset \mathbb{R}$, $c \in \mathbb{R}$ and $f: S \to \mathbb{R}$. Suppose $A \subset S$ s.t $\exists\ \alpha > 0$ for which

$$(A \setminus \{c\}) \cap (c-\alpha, c+\alpha) = (S \setminus \{c\}) \cap (c-\alpha, c+\alpha)$$

(i) c is a cluster point of $A \iff c$ is a cluster point of S

(ii) Suppose c is a cluster point of S then

$$f(x) \to L \text{ as } x \to c \iff f|_A(x) \to L \text{ as } x \to c$$

Proof hint: if $A \cap C = B \cap C$ and $D \subset C$ then $A \cap D = B \cap D$

* **Defn 4.12** Let $f: S \to \mathbb{R}$, $c \in \mathbb{R}$ be a cluster point of $S \cap (c, \infty)$ and the limit of the restriction of f to $S \cap (c, \infty)$ exists then

$$\lim_{x \to c^+} f(x) := \lim_{x \to c} f|_{S \cap (c,\infty)}(x)$$

* **Lemma 4.13** Let $S \subset \mathbb{R}$, c is a cluster point of both $S \cap (c, \infty)$ and $S \cap (-\infty, c)$, $f: S \to \mathbb{R}$, $L \in \mathbb{R}$. Then (i) c is a cluster point of S

(ii) $\lim_{x \to c} f(x) = L \iff \lim_{x \to c^+} f(x) = \lim_{x \to c^-} f(x) = L$

* **Defn 4.14** Let $S \subset \mathbb{R}$, $c \in S$ and $f: S \to \mathbb{R}$. f is called continuous at c if $\forall\ \epsilon > 0\ \exists$ an $\delta > 0$ s.t if $x \in S$ and $|x - c| < \delta$ then $|f(x) - f(c)| < \epsilon$

* **Defn 4.15** $f: S \to \mathbb{R}$ is called continuous if it continuous $\forall c \in S$

* **Lemma 4.16** Let $S \subset \mathbb{R}$, $f: S \to \mathbb{R}$, $c \in S$. then

(i) if c is <u>not</u> a cluster point of S then f is continuous at c.

* (ii) if c is a cluster point of S, then

$$f \text{ is continuous at } c \iff \lim_{x \to c} f(x) = f(c)$$

(iii) f is continuous at $c \iff \forall\ \{x_n\}_{n=1}^{\infty}$ where $x_n \in S$ and $\lim_{n \to \infty} x_n = c$, the sequence $\{f(x_n)\}_{n=1}^{\infty}$ converges to $f(c)$

Hint: Same logic as 4.5

* Lemma 4.17 Let $f: S \to \mathbb{R}$ and $g: S \to \mathbb{R}$ be continuous functions at $c \in S$. then
  (i) $h: S \to \mathbb{R}$ $\quad h(x) := f(x) + g(x)$ is continuous at c
  (ii) " " " $\quad h(x) := f(x)\, g(x)$ " " " "
  (iii) If $g(x) \neq 0$ $\forall x \in S$ then $h: S \to \mathbb{R}$ $\quad h(x): \dfrac{f(x)}{g(x)}$ is continuous at c.

* Lemma 4.18 Let $A, B \subset \mathbb{R}$, $f: B \to \mathbb{R}$, $g: A \to B$. If g is continuous at $c \in A$ and f is continuous at $g(c)$ then $f \circ g: A \to \mathbb{R}$ $\quad f \cdot g(x) := A \to \mathbb{R}$ is continuous at c.
  Hint: Use 4.16(ii)

* Lemma 4.19 $f: [a,b] \to \mathbb{R}$ is continuous $\Rightarrow$ f is bounded
  Hint: Contrapositive statement.
  Use 3.25, 4.16(ii)

* Theorem 4.20
  Extreme value theorem
  A continuous function $f: [a,b] \to \mathbb{R}$ on a closed and bounded interval $[a,b]$ achieves both an absolute maximum and an absolute minimum on $[a,b]$
  Use 4.19, 3.22, 3.25, 3.12

* Lemma 4.21 Let $f: [a,b] \to \mathbb{R}$ be a continuous function. Suppose $f(a) < 0$ and $f(b) > 0$, then $\exists\, c \in (a,b)$ s.t $f(c) = 0$
  Hint: Similar to the proof of 3.25

* Intermediate Value theorem 4.22 Let $f: [a,b] \to \mathbb{R}$ be a continuous function. Suppose $y \in \mathbb{R}$ is s.t $f(a) < y < (b)$ or $f(a) > y > f(b)$. Then $\exists\, c \in (a,b)$ s.t $f(c) = y$
  Hint: $g(x) := f(x) - y$

* Corollary 4.23 If $f: [a,b] \to \mathbb{R}$ is a continuous function then the direct image $f([a,b])$ is a closed and bounded interval or a single number

* Defn 4.24 Let $S \subset \mathbb{R}$, $f: S \to \mathbb{R}$. f is called uniformly continuously if $\forall \varepsilon > 0$ $\exists$ $\delta > 0$ s.t if $x, y \in S$ s.t $|x - y| < \delta$ then $|f(x) - f(y)| < \varepsilon$.
  independent of

* Theorem 4.25 $f: [a,b] \to \mathbb{R}$ is continuous function $\iff$ $f: [a,b] \to \mathbb{R}$ is uniformly continuous

  Hint continuous $\Rightarrow$ uniform continuity
  Contrapositive statement
  Use 4.5

* Lemma 4.26 $f: S \to \mathbb{R}$ be a uniformly continuous function. Let $\{x_n\}_{n=1}^{\infty}$ be a Cauchy sequence with $\forall x_n \in S$ then $\{f(x_n)\}_{n=1}^{\infty}$ is also Cauchy.

* Lemma 4.27 $f: (a,b) \to \mathbb{R}$ is uniformly continous $\Longleftrightarrow \exists\ L_a = \lim_{x \to a} f(x)$, $L_b = \lim_{x \to b} f(b)$ and $\tilde{f}(x) := \begin{cases} f(x) & \text{if } x \in (a,b) \\ L_a & \text{if } x = a \\ L_b & \text{if } x = b \end{cases}$ is continuous

Use 4.26 and 4.5 to show L_a exits

* Defn 4.28 $f: S \to \mathbb{R}$ is called lipschitz continuous, if $\exists\ K \in \mathbb{R}$ s.t

$$|f(x) - f(y)| < K\,|x-y|$$

* Lemma 4.29 A lipschitz continuous function is uniformly continuous.

Hint for $\varepsilon > 0$ use $\delta = \varepsilon / K$